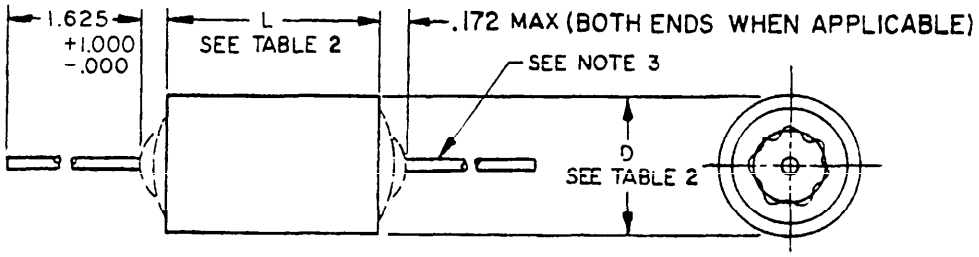
MILITARY SPECIFICATION SHEET

# CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TUBULAR (INSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR39

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this document and the latest issue of MIL-C-19978.

INACTIVE FOR NEW DESIGN
AFTER 20 October 1972.




| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .001 | .03 | .312 | 7.92 |
| .004 | .10 | .400 | 10.16 |
| .025 | .64 | .562 | 14.27 |
| .032 | .81 | 1.000 | 25.40 |
| .172 | 4.37 | 1.562 | 39.67 |
| .235 | 5.97 | 1.625 | 41.28 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned solid wire, .025 (No. 22 AWG) for cases .235 and .312 in diameter; and .032 (No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be +.004, -.001.
4. Capacitors with dimension L of 1.562 or D of .562 and larger, are not intended to be supported by their leads. These capacitors shall be provided with a supplementary means of mounting, such as a wrap-around band.
5. Lead length may be a minimum of 1-inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Style CQR39 capacitors.


FSC 5910
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REQUIREMENTS:

Design and construction:

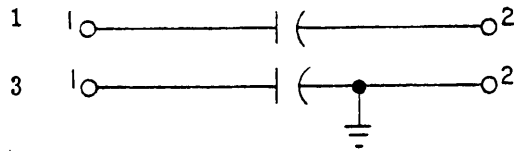
Dimensions and configuration: See figure 1 and table II.

Case material: Nonmagnetic (end seal may be of magnetic material).

Dielectric material: Paper and polyethelene terephthalate(K).

Terminals: Axial-wire lead (see figure 1).

Circuit diagram:




Rated voltage: See table II.

Rated temperature: -65°C to +125°C.

Capacitance (cap.) (nom): See table II.

Capacitance tolerance: K, ±10 percent.

Dissipation factor (DF) (max): 1.0 percent.

Failure rate level: In accordance with MIL-C-19978.

Reliability: In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-C-19978.

Barometric pressure: In accordance with MIL-C-19978.

Test points:

Circuit diagram 1 - Between terminals and case.

Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):

Sleeving: In accordance with MIL-C-19978.

Terminal to terminal: See table I.

Terminal to case: Greater than 10,000 megohms.



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TABLE I. Terminal-to-terminal insulation resistance.

| Capacitance rating | Minimum insulation resistance |
|---|---|
| Characterisitic K | At 25°C |
| 0 to 0.6 microfarad | 25,000 megohms |
| Greater than 0.6 microfarad | 15,000 megohm-microfarads 1/ |
| | At 125°C |
| 0 to 0.08 microfarad | 250 megohms |
| Greater than 0.08 microfarad | 20 megohm-microfarads 1/ |

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Vibration high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Immersion:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

Moisture resistance:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Stability at low and high temperature:

Low temperature:

Test temperature: -65°C +0°, -5°C.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +5°, -0°C.

Capacitance change (max): +10 percent.



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Life:

Capacitance change (max): ±5 percent.

Resistance to soldering heat: In accordance with MIL-C-19978.

TABLE II. Style CQR39 capacitors.

| Part number 1/ | DC voltage rating | Capacitance | Failure rate level | Case dimensions 2/ Characteristic K | | | |
|---|---|---|---|---|---|---|---|
| | | | | Circuit 1 | | Circuit 3 | |
| | | | | L ±0.031 | D +0.015 -0.005 | L ±0.031 | D +0.015 -0.005 |
| | Volts | µF | | Inches | Inches | Inches | Inches |
| CQR39A-KB822K3- | 100 | 0.0082 | L, M, P | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KB103K3- | 100 | 0.010 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KB123K3- | 100 | 0.012 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KB153K3- | 100 | 0.015 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KB393K3- | 100 | 0.039 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KB473K3- | 100 | 0.047 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KB823K3- | 100 | 0.082 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KB104K3- | 100 | 0.10 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KB124K3- | 100 | 0.12 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KB154K3- | 100 | 0.15 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KB184K3- | 100 | 0.18 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KB224K3- | 100 | 0.22 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KB274K3- | 100 | 0.27 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KB334K3- | 100 | 0.33 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KB394K3- | 100 | 0.39 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KB474K3- | 100 | 0.47 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KB564K3- | 100 | 0.56 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KB684K3- | 100 | 0.68 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KB824K3- | 100 | 0.82 | " | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KB105K3- | 100 | 1.0 | " | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KC392K3- | 200 | 0.0039 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KC472K3- | 200 | 0.0047 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KC562K3- | 200 | 0.0056 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KC682K3- | 200 | 0.0068 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KC183K3- | 200 | 0.018 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KC223K3- | 200 | 0.022 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KC273K3- | 200 | 0.027 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KC333K3- | 200 | 0.033 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KC393K3- | 200 | 0.039 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KC473K3- | 200 | 0.047 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KC563K3- | 200 | 0.056 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KC683K3- | 200 | 0.068 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KC823K3- | 200 | 0.082 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KC104K3- | 200 | 0.10 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KC124K3- | 200 | 0.12 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KC154K3- | 200 | 0.15 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KC184K3- | 200 | 0.18 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KC224K3- | 200 | 0.22 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KC274K3- | 200 | 0.27 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KC334K3- | 200 | 0.33 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KC394K3- | 200 | 0.39 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KC474K3- | 200 | 0.47 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KC564K3- | 200 | 0.56 | " | 1.625 | 0.670 | 1.562 | 0.670 |

See footnotes at end of table.



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TABLE II. Style CQR39 capacitors - Continued.

| Part number 1/ | DC voltage rating | Capacitance | Failure rate level | Case dimensions 2/ Characteristic K Circuit 1 L ±0.031 | Circuit 1 D +0.015 -0.005 | Circuit 3 L ±0.031 | Circuit 3 D +0.015 -0.005 |
|---|---|---|---|---|---|---|---|
| | Volts | µF | | Inches | Inches | Inches | Inches |
| CQR39A-KC684K3- | 200 | 0.68 | L, M, P | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KC824K3- | 200 | 0.82 | " | 2.125 | 0.750 | 2.062 | 0.750 |
| CQR39A-KC105K3- | 200 | 1.0 | " | 2.125 | 0.750 | 2.062 | 0.750 |
| CQR39A-KE272K3- | 400 | 0.0027 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KE332K3- | 400 | 0.0033 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KE123K3- | 400 | 0.012 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KE153K3- | 400 | 0.015 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KE273K3- | 400 | 0.027 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KE333K3- | 400 | 0.033 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KE393K3- | 400 | 0.039 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KE473K3- | 400 | 0.047 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KE563K3- | 400 | 0.056 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KE683K3- | 400 | 0.068 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KE823K3- | 400 | 0.082 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KE104K3- | 400 | 0.10 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KE124K3- | 400 | 0.12 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KE154K3- | 400 | 0.15 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KE184K3- | 400 | 0.18 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KE224K3- | 400 | 0.22 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KE274K3- | 400 | 0.27 | " | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KE334K3- | 400 | 0.33 | " | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KE394K3- | 400 | 0.39 | " | 2.125 | 0.750 | 2.062 | 0.750 |
| CQR39A-KE474K3- | 400 | 0.47 | " | 2.125 | 0.750 | 2.062 | 0.750 |
| CQR39A-KF102K3- | 600 | 0.0010 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KF122K3- | 600 | 0.0012 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KF152K3- | 600 | 0.0015 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KF182K3- | 600 | 0.0018 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KF222K3- | 600 | 0.0022 | " | 0.750 | 0.235 | 0.688 | 0.235 |
| CQR39A-KF272K3- | 600 | 0.0027 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KF332K3- | 600 | 0.0033 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KF392K3- | 600 | 0.0039 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KF472K3- | 600 | 0.0047 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KF562K3- | 600 | 0.0056 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KF682K3- | 600 | 0.0068 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KF822K3- | 600 | 0.0082 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KF103K3- | 600 | 0.010 | " | 0.875 | 0.312 | 0.812 | 0.312 |
| CQR39A-KF123K3- | 600 | 0.012 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KF153K3- | 600 | 0.015 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KF183K3- | 600 | 0.018 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KF223K3- | 600 | 0.022 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KF273K3- | 600 | 0.027 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KF333K3- | 600 | 0.033 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KF393K3- | 600 | 0.039 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KF473K3- | 600 | 0.047 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KF563K3- | 600 | 0.056 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KF683K3- | 600 | 0.068 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KF823K3- | 600 | 0.082 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KF104K3- | 600 | 0.10 | " | 1.375 | 0.562 | 1.312 | 0.562 |

See footnotes at end of table.



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TABLE II. Style CQR39 capacitors - Continued.

| Part number 1/ | DC voltage rating | Capacitance | Failure rate level | Case dimensions 2/ Characteristic K | | | |
|---|---|---|---|---|---|---|---|
| | | | | Circuit 1 | | Circuit 3 | |
| | | | | L ±0.031 | D +0.015 -0.005 | L ±0.031 | D +0.015 -0.005 |
| | Volts | µF | | Inches | Inches | Inches | Inches |
| CQR39A-KF124K3- | 600 | 0.12 | L, M, P | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KF154K3- | 600 | 0.15 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KF184K3- | 600 | 0.18 | " | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KF224K3- | 600 | 0.22 | " | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KF274K3- | 600 | 0.27 | " | 2.125 | 0.750 | 2.062 | 0.750 |
| CQR39A-KF334K3- | 600 | 0.33 | " | 2.125 | 0.750 | 2.062 | 0.750 |
| CQR39A-KF394K3- | 600 | 0.39 | " | 2.375 | 0.750 | 2.312 | 0.750 |
| CQR39A-KF474K3- | 600 | 0.47 | " | 2.375 | 0.750 | 2.312 | 0.750 |
| CQR39A-KG102K3- | 1,000 | 0.001 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG122K3- | 1,000 | 0.0012 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG152K3- | 1,000 | 0.0015 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG182K3- | 1,000 | 0.0018 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG222K3- | 1,000 | 0.0022 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG272K3- | 1,000 | 0.0027 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG332K3- | 1,000 | 0.0033 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG392K3- | 1,000 | 0.0039 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG472K3- | 1,000 | 0.0047 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG562K3- | 1,000 | 0.0056 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG682K3- | 1,000 | 0.0068 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG822K3- | 1,000 | 0.0082 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG103K3- | 1,000 | 0.01 | " | 0.875 | 0.400 | 0.812 | 0.400 |
| CQR39A-KG123K3- | 1,000 | 0.012 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KG153K3- | 1,000 | 0.015 | " | 1.125 | 0.400 | 1.062 | 0.400 |
| CQR39A-KG183K3- | 1,000 | 0.018 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KG223K3- | 1,000 | 0.022 | " | 1.375 | 0.400 | 1.312 | 0.400 |
| CQR39A-KG273K3- | 1,000 | 0.027 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KG333K3- | 1,000 | 0.033 | " | 1.125 | 0.562 | 1.062 | 0.562 |
| CQR39A-KG393K3- | 1,000 | 0.039 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KG473K3- | 1,000 | 0.047 | " | 1.375 | 0.562 | 1.312 | 0.562 |
| CQR39A-KG563K3- | 1,000 | 0.056 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KG683K3- | 1,000 | 0.068 | " | 1.625 | 0.562 | 1.562 | 0.562 |
| CQR39A-KG823K3- | 1,000 | 0.082 | " | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KG104K3- | 1,000 | 0.10 | " | 1.625 | 0.670 | 1.562 | 0.670 |
| CQR39A-KG124K3- | 1,000 | 0.12 | " | 1.875 | 0.670 | 1.812 | 0.670 |
| CQR39A-KG154K3- | 1,000 | 0.15 | " | 1.875 | 0.670 | 1.812 | 0.670 |
| CQR39A-KG184K3- | 1,000 | 0.18 | " | 2.125 | 0.750 | 2.062 | 0.750 |
| CQR39A-KG224K3- | 1,000 | 0.22 | " | 2.125 | 0.750 | 2.062 | 0.750 |
| CQR39A-KG274K3- | 1,000 | 0.27 | " | 2.125 | 1.000 | 2.062 | 1.000 |
| CQR39A-KG334K3- | 1,000 | 0.33 | " | 2.125 | 1.000 | 2.062 | 1.000 |
| CQR39A-KG394K3- | 1,000 | 0.39 | " | 2.375 | 1.000 | 2.312 | 1.000 |
| CQR39A-KG474K3- | 1,000 | 0.47 | " | 2.375 | 1.000 | 2.312 | 1.000 |

1/ Complete part number shall include additional symbols to indicate circuit, and failure rate level, as applicable.

2/ Dimensions are for basic case; sleeving shall extend 0.016 inch minimum, 0.062 inch maximum, beyond each end of the capacitor body; however, if a shrink-fitted insulation is used for the sleeving, it shall lap over the ends of the capacitor body. Add 0.047 inch maximum to the nominal for capacitor diameter.



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TABLE III. Millimeters equivalents of decimal inches.

| Inches | MM | Inches | MM | Inches | MM |
|---|---|---|---|---|---|
| 0.005 | 0.13 | 0.562 | 14.27 | 1.312 | 33.32 |
| 0.015 | 0.38 | 0.670 | 17.02 | 1.375 | 34.93 |
| 0.016 | 0.41 | 0.688 | 17.48 | 1.562 | 39.67 |
| 0.031 | 0.79 | 0.750 | 19.05 | 1.625 | 41.28 |
| 0.047 | 1.19 | 0.812 | 20.62 | 1.875 | 47.63 |
| 0.062 | 1.57 | 0.875 | 22.23 | 2.062 | 52.37 |
| 0.235 | 5.97 | 1.000 | 25.40 | 2.125 | 53.98 |
| 0.312 | 7.92 | 1.062 | 26.97 | 2.312 | 58.72 |
| 0.400 | 10.16 | 1.125 | 28.58 | 2.375 | 60.33 |

TABLE IV. Substitutability data. 1/

| From | To | From | To |
|---|---|---|---|
| MIL-C-25/1K (CP08 and CP09) | MIL-C-19978/16 (CQR39) | MIL-C-25/1K (CP08 and CP09) | MIL-C-19978/16 (CQR39) |
| CP--A-KB822K- | CQR39A-KB822K3- | CP--A-KC224K- | CQR39A-KC224K3- |
| CP--A-KB103K- | CQR39A-KB103K3- | CP--A-KC274K- | CQR39A-KC274K3- |
| CP--A-KB123K- | CQR39A-KB123K3- | CP--A-KC334K- | CQR39A-KC334K3- |
| CP--A-KB153K- | CQR39A-KB153K3- | CP--A-KC394K- | CQR39A-KC394K3- |
| CP--A-KB393K- | CQR39A-KB393K3- | CP--A-KC474K- | CQR39A-KC474K3- |
| CP--A-KB473K- | CQR39A-KB473K3- | CP--A-KC564K- | CQR39A-KC564K3- |
| CP--A-KB823K- | CQR39A-KB823K3- | CP--A-KC684K- | CQR39A-KC684K3- |
| CP--A-KB104K- | CQR39A-KB104K3- | CP--A-KC824K- | CQR39A-KC824K3- |
| CP--A-KB124K- | CQR39A-KB124K3- | CP--A-KC105K- | CQR39A-KC105K3- |
| CP--A-KB154K- | CQR39A-KB154K3- | CP--A-KE272K- | CQR39A-KE272K3- |
| CP--A-KB184K- | CQR39A-KB184K3- | CP--A-KE332K- | CQR39A-KE332K3- |
| CP--A-KB224K- | CQR39A-KB224K3- | CP--A-KE123K- | CQR39A-KE123K3- |
| CP--A-KB274K- | CQR39A-KB274K3- | CP--A-KE153K- | CQR39A-KE153K3- |
| CP--A-KB334K- | CQR39A-KB334K3- | CP--A-KE273K- | CQR39A-KE273K3- |
| CP--A-KB394K- | CQR39A-KB394K3- | CP--A-KE333K- | CQR39A-KE333K3- |
| CP--A-KB474K- | CQR39A-KB474K3- | CP--A-KE393K- | CQR39A-KE393K3- |
| CP--A-KB564K- | CQR39A-KB564K3- | CP--A-KE473K- | CQR39A-KE473K3- |
| CP--A-KB684K- | CQR39A-KB684K3- | CP--A-KE563K- | CQR39A-KE563K3- |
| CP--A-KB824K- | CQR39A-KB824K3- | CP--A-KE683K- | CQR39A-KE683K3- |
| CP--A-KB105K- | CQR39A-KB105K3- | CP--A-KE823K- | CQR39A-KE823K3- |
| CP--A-KC392K- | CQR39A-KC392K3- | CP--A-KE104K- | CQR39A-KE104K3- |
| CP--A-KC472K- | CQR39A-KC472K3- | CP--A-KE124K- | CQR39A-KE124K3- |
| CP--A-KC562K- | CQR39A-KC562K3- | CP--A-KE154K- | CQR39A-KE154K3- |
| CP--A-KC682K- | CQR39A-KC682K3- | CP--A-KE184K- | CQR39A-KE184K3- |
| CP--A-KC183K- | CQR39A-KC183K3- | CP--A-KE224K- | CQR39A-KE224K3- |
| CP--A-KC223K- | CQR39A-KC223K3- | CP--A-KE274K- | CQR39A-KE274K3- |
| CP--A-KC273K- | CQR39A-KC273K3- | CP--A-KE334K- | CQR39A-KE334K3- |
| CP--A-KC333K- | CQR39A-KC333K3- | CP--A-KE394K- | CQR39A-KE394K3- |
| CP--A-KC393K- | CQR39A-KC393K3- | CP--A-KE474K- | CQR39A-KE474K3- |
| CP--A-KC473K- | CQR39A-KC473K3- | CP--A-KF102K- | CQR39A-KF102K3- |
| CP--A-KC563K- | CQR39A-KC563K3- | CP--A-KF122K- | CQR39A-KF122K3- |
| CP--A-KC683K- | CQR39A-KC683K3- | CP--A-KF152K- | CQR39A-KF152K3- |
| CP--A-KC823K- | CQR39A-KC823K3- | CP--A-KF182K- | CQR39A-KF182K3- |
| CP--A-KC104K- | CQR39A-KC104K3- | CP--A-KF222K- | CQR39A-KF222K3- |
| CP--A-KC124K- | CQR39A-KC124K3- | CP--A-KF272K- | CQR39A-KF272K3- |
| CP--A-KC154K- | CQR39A-KC154K3- | CP--A-KF332K- | CQR39A-KF332K3- |
| CP--A-KC184K- | CQR39A-KC184K3- | CP--A-KF392K- | CQR39A-KF392K3- |

See footnote at end of table.



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TABLE IV. Substitutability data 1/ - Continued.

| From | To | From | To |
|---|---|---|---|
| MIL-C-25/1K (CP08 and CP09) | MIL-C-19978/16 (CQR39) | MIL-C-25/1K (CP08 and CP09) | MIL-C-19978/16 (CQR39) |
| CP--A-KF472K- | CQR39A-KF472K3- | CP--A-KG222K- | CQR39A-KG222K3- |
| CP--A-KF562K- | CQR39A-KF562K3- | CP--A-KG272K- | CQR39A-KG272K3- |
| CP--A-KF682K- | CQR39A-KF682K3- | CP--A-KG332K- | CQR39A-KG332K3- |
| CP--A-KF822K- | CQR39A-KF822K3- | CP--A-KG392K- | CQR39A-KG392K3- |
| CP--A-KF103K- | CQR39A-KF103K3- | CP--A-KG472K- | CQR39A-KG472K3- |
| CP--A-KF123K- | CQR39A-KF123K3- | CP--A-KG562K- | CQR39A-KG562K3- |
| CP--A-KF153K- | CQR39A-KF153K3- | CP--A-KG682K- | CQR39A-KG682K3- |
| CP--A-KF183K- | CQR39A-KF183K3- | CP--A-KG822K- | CQR39A-KG822K3- |
| CP--A-KF223K- | CQR39A-KF223K3- | CP--A-KG103K- | CQR39A-KG103K3- |
| CP--A-KF273K- | CQR39A-KF273K3- | CP--A-KG123K- | CQR39A-KG123K3- |
| CP--A-KF333K- | CQR39A-KF333K3- | CP--A-KG153K- | CQR39A-KG153K3- |
| CP--A-KF393K- | CQR39A-KF393K3- | CP--A-KG183K- | CQR39A-KG183K3- |
| CP--A-KF473K- | CQR39A-KF473K3- | CP--A-KG223K- | CQR39A-KG223K3- |
| CP--A-KF563K- | CQR39A-KF563K3- | CP--A-KG273K- | CQR39A-KG273K3- |
| CP--A-KF683K- | CQR39A-KF683K3- | CP--A-KG333K- | CQR39A-KG333K3- |
| CP--A-KF823K- | CQR39A-KF823K3- | CP--A-KG393K- | CQR39A-KG393K3- |
| CP--A-KF104K- | CQR39A-KF104K3- | CP--A-KG473K- | CQR39A-KG473K3- |
| CP--A-KF124K- | CQR39A-KF124K3- | CP--A-KG563K- | CQR39A-KG563K3- |
| CP--A-KF154K- | CQR39A-KF154K3- | CP--A-KG683K- | CQR39A-KG683K3- |
| CP--A-KF184K- | CQR39A-KF184K3- | CP--A-KG823K- | CQR39A-KG823K3- |
| CP--A-KF224K- | CQR39A-KF224K3- | CP--A-KG104K- | CQR39A-KG104K3- |
| CP--A-KF274K- | CQR39A-KF274K3- | CP--A-KG124K- | CQR39A-KG124K3- |
| CP--A-KF334K- | CQR39A-KF334K3- | CP--A-KG154K- | CQR39A-KG154K3- |
| CP--A-KF394K- | CQR39A-KF394K3- | CP--A-KG184K- | CQR39A-KG184K3- |
| CP--A-KF474K- | CQR39A-KF474K3- | CP--A-KG224K- | CQR39A-KG224K3- |
| CP--A-KG102K- | CQR39A-KG102K3- | CP--A-KG274K- | CQR39A-KG274K3- |
| CP--A-KG122K- | CQR39A-KG122K3- | CP--A-KG334K- | CQR39A-KG334K3- |
| CP--A-KG152K- | CQR39A-KG152K3- | CP--A-KG394K- | CQR39A-KG394K3- |
| CP--A-KG182K- | CQR39A-KG182K3- | CP--A-KG474K- | CQR39A-KG474K3- |

1/ NOTE: Style CQR39 capacitors are substitutable for styles CP08 and CP09 capacitors formerly covered in MIL-C-25/1K.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC, OS, SH

Preparing activity:
Air Force - 85

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